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Optimization and improvement of fake news detection using deep learning approaches for societal benefit

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A B S T R A C T

Fake news is a topic that has been discussed for quite some time. Prior to the internet era, it was mostly distributed through yellow journalism, with a focus on sensational news such as crime, rumours, accidents, and amusing news. To rescue the life of people from these fake news propagation, detection of fake news at an early stage becomes the most crucial step. People unknowingly propagate fake news and become a part of fake news propagation. While the original fake news propagators are the one with their aim to target innocent people for spreading the fake news. To stop this series of events, fake news detection and its pattern of propagation becomes very essential to society and the government. Various techniques exist to detect fake news in social media, among which neural networks have shown effective results. For this research, a deep learning based approach has been used to differentiate false news from the original ones. A LSTM neural network has been used to build the proposed model. Besides the neural network, a gloVe word embedding has been used for vector representation of textual words. Also, for feature extraction or vectorization, tokenization technique has been used. N-grams concept is used to enhance the proposed model. The comparative analysis of multiple fake news detection techniques is analysed. The results of proposed model have been evaluated using accuracy metrics. The model outperformed by achieving 99.88% of accuracy.

1. Introduction

On this technological planet, there are plenty of growing technologies that assist in understanding human behaviour. Earlier it was the age when human-to-machine interaction was just imagined as a hallucination. Around our planet, a huge heavenly body, there exists a civilization of living beings and a relationship that links us to the same land. It bridges us with the connection of belonging to the same planet and builds a huge society to live in. To develop this society there are a group of people with the authority to govern a certain region. Society is surrounded by numerous aspects mainly the humans, the environment, multiple industries that could serve for the betterment of people. To supervise these activities government of that particular state develops certain rules that need to be followed by society. To assist the Government, various emerging technologies could help Government to build secure policies. Artificial intelligence is one of the overpowering technology which was coined by John McCarthy in 1955. Later on machine learning, deep learning, natural language processing, neural networks, predictive analysis were discovered. With the emerging technologies, there has been considerable development in each domain. Hence, even the government has been incorporating these technologies in every possible policy that are for the betterment of people.

Artificial intelligence is one of the emerging technologies that has changed the sight of looking at business issues (Palanivelu & Vasanthi, 2020; Ruiz-Real, Uribe-Toril, Torres, & De Pablo, 2021). A rising number of companies are turning to advanced analytics and machine learning to solve problems. With this evolution in the era of artificial intelligence, natural language processing (NLP) unfolds abundant opportunities for businesses that are interested in understanding human sentiments through the existing data (Kang, Cai, Tan, Huang, & Liu, 2020). NLP can function with all types of natural and social communication, including audio, video and text. To recognize various multiple and useful patterns and trends in the textual dataset, text mining has been assisting to do so. In today's market settings, strategic usage of NLP enables businesses to acquire a competitive advantage over other competitors. Artificial intelligence and natural language processing can help combat the huge unstructured data of multiple domains including healthcare, education, fake news, business sectors, security and trust, as well as opinions from the community in the government sectors. Natural language processing enables human-to-machine interaction more effectively which in turn enhances the decision-making and overall efficiency of the business (Bahja, 2020). Natural language refers to how individuals interact, which includes speech, text, and emotions. Speaking as a mode of communication engages people more than writing (text) (Eisenstein, 2019).

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The amount of data produced by humans and machines nowadays significantly outnumbers humans' potential to withstand, evaluate, and make complicated decisions based on that data. One of the major challenges that businesses go through is to analyze human behaviour. Business intelligence is used to provide an easy approach to integrate chatbots to enhance business developments (Vashisht & Dharia, 2020). The authors in this paper (Yuan et al., 2021) aims at analysing business reviews based on the past reviewers to make recommendations of future ideas and increment the profit growth of the business. This is where natural language processing is used to assist business needs and increment the business profit as well as provide customized services to their customers. These online reviews also affect business decisions as well as e-commerce and e-governance. Hence, Governments wish to enhance their policies and services and shrink the gap between the citizens and government. At certain times, even society could mislead the internet against the original contents which are not supported by the government. Governments can also use NLP approaches to improve feedback analysis, regulatory compliance, and policy analysis, all of which would be beneficial to society (Kang et al., 2020). The findings reveal that many researchers looked into trust as a key component in e-government adoption in underdeveloped countries, rather than vital issues like security and knowledge received (Mustaf, Ibrahim, & Mohammed, 2020).

The security and trust issues in e-government also give rise to developing fake news that might in turn harm society as well as the government also. Fake news is such an astound segment of social media that its influences and its expansion can lead to dreadful incidents. The authors of this paper (Cheng & Chen, 2020) understands the public support and important impacts of consumer engagement, self-efficacy, and impulsiveness on presumed fake news. Several circumstances have had an impact on society, resulting in not only a contribution of advantages but also of danger in some situations. Because of the danger, it poses to society, the fake news epidemic is regarded as one of the most hotly debated topics today. The findings of the authors in the paper (García, García, Prieto, Moreno Guerrero, & Rodríguez Jiménez, 2020) demonstrated the current influence of the fake news subject, which is seen as an important trend of relevance to many scientific disciplines across the world. The findings revealed that the publications not only rely on examining the authenticity but also begin to pursue a new line of enquiry aimed at educational literacy and limiting the use of this sort of news over the internet.

Fake news tends to spread faster and is unrecognized. If fake news is once prominent in print, has become more prevalent with the emergence of social media, particularly the Facebook News Feed. The proliferation of fake news has been associated with political extremism, post-truth politics. The authors of this paper (Vosoughi, Roy, & Aral, 2018) tried to understand and analyze the impact of fake news on society by examining a data set of rumour cascades on Twitter from 2006 to 2017 to better understand how fake news spreads. The authors also identified the spread of fake news in bulk. A total of 3 million people circulated about 126,000 rumours. False news reached a larger audience than true news.

Hence, the purpose of uncovering fake news must be for the benefit of society as well as for the support of the government. Rather than trying to cover the spread of fake news post-expansion, such an administrative system is required that could combat the after-effects of the spread of fake news and save society from becoming a mysterious part of spreading fake news. This research aims to identify and understand the influence of fake news on the government as well as on society. Thus, fake news propagation has been in the buzz since a long time ago. For the welfare of society, it is the prime responsibility to detect fake news. A result of the hugely growing social media data and emerging technologies make it essential and helpful to realize fake news before it starts to spread in numbers.

Apart from shedding light on the significance of fake news detection, an approach has been built. To improve the findings of fake news, a deep learning-based LSTM neural network model has been proposed. The proposed model assists in detecting fake news and real news. Firstly, the concept of stopwords has been used before training the model. For feature extraction or vectorization, a method of tokenization has been approached that a lots tokens to the word embeddings. For word embeddings, a GloVe concept has been followed that represents each word to vector form. After this, the text passes through the main LSTM neural network where it goes under different layers present in the architecture. Ultimately, the model is then set to recognize between true news and false news.

The next elements of the paper are congregated as follows. The following Section 2, gives insights on the aim of doing this research and collection of research work by multiple authors concerning fake news, fake news propagation, its impact on society and techniques that exists to detect fake news. Section 3, unfolds the sceptical incidents of the propagation of fake news and its knock-on the society. Section 4, sheds light on the proposed model, information about the dataset and the building blocks required to propose the fake news detection model. The results of the proposed model and its performance evaluation is carried out in Section 5. The next Section 6, describes the discussion of theoretical contributions and implications to practise. Eventually, Section 7 describes the conclusion of the findings and proposed model.

2. Research goal and literature survey

Fake news is a subject that has been around for a long time. It was mostly transmitted through yellow journalism before the internet era, with a concentration on spectacular news such as crime, rumours, accidents, and hilarious news (Stein-Smith, 2017). Because of the character that social media consists of, it is simple to propagate fake news, as a user may share fake news to friends, who can then transmit it to their friends, and this cycle goes on. Comments on fake news can at various times instigate its 'reliability,' directing it towards more fake news and increase its spread by double the original time speed (Albright, 2017). Our paper's fundamental theme is concerned with the objective of identifying if the contents of social media are honest or fake. The significance of social learning for various operational decisions have risen in popularity.

Among the work of operational decisions, the presence of social learning can reduce the quality of new experiences. The related work suggests the impact of fake news on society. Fake news can have a wide range of effects, from being just bothersome to influencing and misleading entire populations or even governments (Feldman, Papanastasiou, & Segev, 2019). Few of the approaches for detecting fake news that is currently accessible involves knowledge-based approach, language approach, machine learning approach, hybrid approach and topic-agnostic approach (De Beer & Matthee, 2020). As a coin has two sides, similarly technology and social media walk hand in hand and have their profits that come along with certain pitfalls. Facebook, Instagram, YouTube, and Twitter have become strong platforms for providing news and entertainment, thanks to the growing use of mobile devices and easy Wi-Fi connection to 3G/4G networks. With the positive impact of social media and technology on society, each one has its limitations or extremities. A few of the limitations and advantages of using social media and an analysis of various techniques that could help in detecting fake news is mentioned here (Bondielli & Marcelloni, 2019). Fake news propagators fabricate into the internet community by providing false information. Machine learning techniques have also been utilized to build a predictive fake news model. By analysing the numerous feature of fake news, the authors approach to detect the fake news by analysing those features (Singh, Ghosh, & Sonagara, 2021). A set of tools or applications are

investigated through which the fake news tends to spread. They involve the most commonly used applications that are Twitter, WhatsApp and Facebook. These are the main platforms where innocent people come across fake news propagators (Igwebuike & Chimuanya, 2020). Additionally, a computational linguistic approach is also followed to the language of users on US twitter that promotes falsehood (Li & Su, 2020).

The spread of populist rhetoric, divisive views, and spiteful opinions is facilitated by the spread of misinformation along with the appropriation of cultural symbolism, and the active participation of news media in propagating the information. The authors unfold the impact after U.S. presidential election in 2016. The expansion of members of society and its vision is unique in its origins and is a direct outcome of a highly radical and sceptical public spending more time in social networks. Authors analysed how the presidential news started to spread and then gradually got expanded. The author said about the media literacies; that as a prominent reaction mechanism for cultivating more critical media consumers must be realigned to answer to the public that maintains the parameter of hostility and distrust (Mihailidis & Viotty, 2017).

The creation and consumption of news and information in our society are evolving in this new era of the internet and multiple platforms of social media. Traditional media's quick metamorphosis into online portals has become the latest craze (Shu, Wang, Lee, & Liu, 2020). Online social media has standardized news production and dissemination, but it has also turned into a seedbed, for misleading and fake news leading to misinformation (Nagi, 2018). The President and Prime Minister of India as well as the United States, respectively, use Twitter to communicate with their followers and voters. As a result, the direct connection of politicians and officials with citizens via social media has a significant impact on how governments around the world operate. A traditional and analytical approach is followed to understand and analyse the present issues relating to fake news. Also, the authors find about the negative impact of fake news propagation on the society (Nagi, 2018). The appearance of fake news on social media platforms is analyzed. The author goes on to examine the current scenario and also evaluates the findings of the 2018 Media Literacy Index, which examines countries' tolerance to the aforementioned harmful occurrences. Prevention options to fake news have been analyzed, as well as education in this area and promoting media literacy. The study offers a theoretical perspective on an issue that has resurfaced in recent years in society and focuses on safety precautions that need to be taken. With the centre of attention to take efforts in the fields of education and media literacy for the general public, it is equally essential to knowledge the society and prevents the civilization with necessary measures to save people from becoming the culprit to growing fake news (Hossová et al., 2018).

The use of natural language processing enables us to construct such a model that could efficiently recognize between fake news and real news. To tackle these flaws of fake news, the authors approach natural language processing (NLP) and suggest that fact-checking be combined with linguistic features analysis to truly distinguish false news from true news (Zhou, Guan, Bhat, & Hsu, 2019). To detect fake news a ConvNet-RNN hybrid approach along with the LSTM model was used to classify fake news and real news (Ma & Tan, 2021). In this paper, the authors decided to use a random forest algorithm along with NLP to differentiate between trustworthy news and untrustworthy news (Vijay, Basha, & Nehru, 2021). A semantic false news detection system is utilized based on relational features taken directly from the text, such as sentiment, entities, and facts. It used short texts with varying degrees of truth to demonstrate that adding semantic characteristics improves accuracy dramatically (Braşoveanu & Andonie, 2020). Existing detection approaches rely on either news content or social context, as well as specific user-based data. For fake news identification, the substance of the newspaper article and the presence of echo chambers (communities of social media-based users who share the same viewpoints) in the social network are considered by the researchers (Kaliyar, Goswami, & Narang, 2021a). To detect false news, the content of both news pieces and the prevalence of echo chambers are focused on the social network. Because of their unsupervised nature, standard factorization methods for detecting false news are generally used with traditional machine learning models (Kaliyar, Goswami, & Narang, 2021b). The authors in this article (Kaliyar, Goswami, & Narang, 2021c) propose a FakeBERT deep learning strategy based on BERT (Bidirectional Encoder Representations from Transformers) by merging several parallel blocks of a single-layer deep Convolutional Neural Network (CNN) with variable kernel sizes and screens with the BERT. A novel approach for detecting fraudulent news was proposed, which incorporates optimization methods (Ozbay & Alatas, 2021). For the first time, an enhanced Salp Swarm Optimization (SSO) was proposed, based on a nonlinear decreasing coefficient and oscillating inertia weight, to identify the best optimum solution for fake news identification. Textual and visual modules are created to examine the performances over multi-modal datasets to enhance ongoing research on false news identification using CNN models. Using layers of convolutions, latent patterns are leveraged and found in text and images. The authors present a multi-modal Coupled ConvNet architecture that effectively classifies online news based on its textual and visual content by fusing both data modules (Raj & Meel, 2021).

Besides the propagation of fake news and its impact on society, there have been various techniques that exist to discriminate between false news and real news. A Graph-aware Co-Attention Networks (GCAN) is created, which is a unique neural network-based model to predict if a particular tweet is fake or real (Lu & Li, 2020). An autonomous fake news detection methodology is introduced in the Chrome environment that can detect fake news on Facebook. Also, the authors employ several variables linked with a Facebook account, as well as some news article features, to assess the account's activity using deep learning (Sahoo & Gupta, 2021). Here machine learning techniques are used to detect fake news. The experiments employ three common methods: Naive Bayes, Neural Networks, and Support Vector Machines (Aphiwongsophon & Chongstitvatana, 2018). There is a significant need to create methods for detecting false news based on news content. A theory-driven model for detecting fake news is proposed by authors in this article (Zhou, Jain, Phoha, & Zafarani, 2020). The technique looks into news content on several levels: At the lexicon, syntax, semantic, and discourse levels. For false news classification, the author of this paper (Nasir, Khan, & Varlamis, 2021) introduces a novel hybrid deep learning model that blends convolutional and recurrent neural networks. An approach is made for converting the GloVe (Global Vectors for Word Representation) model, a common technique for word embeddings, to a novel method called GloVeNoR, which evaluates node embeddings in a graph and creates a corpus using second-order random walks, is explored in this study (Kulkarni, Katariya, & Potika, 2020). The use of a glove as a simple and low-cost human-machine interface HMI can quickly follow the motions of human fingers, resulting in a simple human-machine interface communication medium. It has significant potential to perform complicated gesture recognition with the minimalist-designed glove for complete control in both real and virtual space when numerous textile sensors and correct machine learning methods are combined. Various gesture recognition activities are done in real-time by using gestures to provide very effective virtual reality/augmented reality (VR/AR) controls, such as pistol shooting, baseball throwing, and flower arrangement, which is all possible due to machine learning technology (Wen et al., 2020).

3. Theory to fake news

The Nigerian society had been facing threats socially, economically as well as politically during the times of Nigeria's 2019 general elections. The culprit to this threat to the Nigerian public was the spread of fake news during the period of election. Fake news and the accompanying concept of post-truth, according to this research, may continue to represent a threat to the Nigerian polity unless robust steps would be adopted.

To effectively suppress the effects of fake news and post-truth phenomena, a threefold involvement model incorporating these major stakeholders - the government, legislators, and the general people - should be evolved and then administered to the letter (Pate et al., 2019). The post-truth has a significant impact on the media involving politics. There occurs a difference in the effects of fake news and the impact of it after post-truth (Corner, 2017). Twitter is becoming more widely utilised in the political sphere as a means of disseminating thoughts and feedback. Donald Trump being one of the nominees for the 2016 US Presidential elections used Twitter as a means to connect to the public and went on even after the victory. Trump's use of Twitter had attracted criticism for his speech on a variety of matters which included comments on Hillary Clinton, as well as the number of the audience at his inauguration, prior Obama administration policies, immigration and foreign relations. Among the most noticeable aspects of Trump's Twitter use has been his frequent sarcasm of the mass media by the use of terms like "fake news" and "dishonest media." These terms gave rise to the public regarding the trust of media reports. Given the objective reality, it could be claimed that Trump, like the media, is the culprit for the transmission of misinformation (Ross & Rivers, 2018).

Fake news has even conquered the current situation of the COVID-19 pandemic. And this time among various social media platforms, Whatsapp has trapped people into the circulation of fake news. Several films claiming to portray sights from China like images that show sights of victims lying on streets - are circulating on WhatsApp to demonstrate the impact of coronavirus. Several of these videos have been exposed as outdated footage, mock drills, or even sequences from movies by fact-checking organisations. Also, fact-checkers have questioned the government's recommendation, which recommends plants and homoeopathic treatments as remedies to prevent COVID-19. This enhanced the level of spread of fake news in such a way that people had started believing in these rumours. The fact-checking organization reported on a man in southern India who committed suicide after seeing multiple videos of coronavirus and assuming that he also might have got infected. This empowered the sight of government and immediate actions had to be taken where 3000 people were in the surveillance. It was after this when the central government had to take a crucial step to combat the spread of fake news. To combat disinformation, the central government then ordered telecom providers to provide an audio recording clarifying coronavirus as a caller tune for mobile phone customers Purohit (2020). Also, health worker had been arrested to propagate the fake news (Saez-Trumper, 2019).

4. Methodology

In correspondence with this research, our goal is to train our model to correctly predict whether a piece of news is true or false. Firstly, we aim to build a model that could beneficially help the government as well as our society. Secondly, our focus is to minimize the spread of fake news through early detection so that it can be stopped from further expansion. Next, we set our sights on saving society from transmitting fake news unknowingly so that it can be out of reach of the propagating community.

4.1. Dataset description

To differentiate the real news from fake news, the data has been collected from two different origins. The fake and real news dataset is taken from kaggle.com (Ahmed, Traore, & Saad, 2018). The dataset in all contains about 40,000 articles among which includes both fake news as well as real news. The false news data and actual news data is separated into two different datasets, each with approximately 20,000 articles. Whereas another dataset is the glove Twitter data which is a pre-trained dataset and is identified by the authors in this article (Pennington, Socher, & Manning, 2014). The dataset contains four columns that are, title, the main text, subject and date.

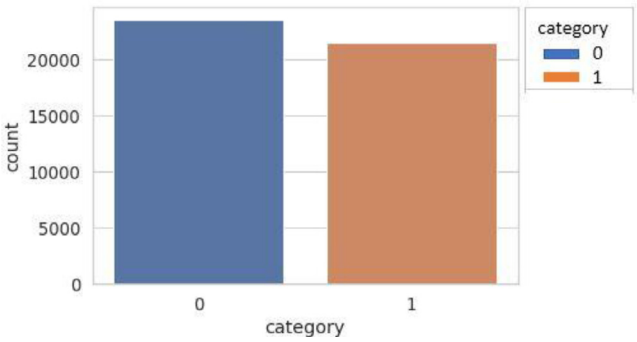

Fig. 1. News articles in correspondence with fake and real category.

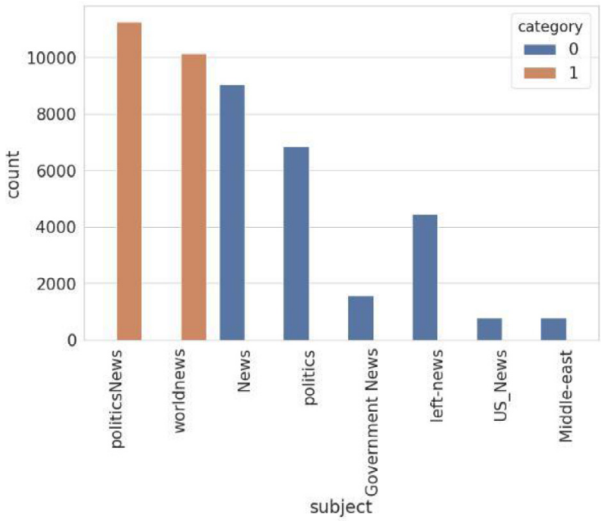

Fig. 2. Topics trending in each category of fake news and real news.

4.2. Data visualization and preprocessing

The dataset is characterized into two classes, one labelled as a true category and the second class labelled as a false category. Data visualisation helps us comprehend what relative data means by displaying data in a visual context, such as maps or graphs. This makes it easier to spot trends, patterns, and outliers in large data sets by making the data more natural to analyse for the human mind. The dataset is classified into two categories that are fake news and original news. The first category is true news category represented by class '1' and second category is fake news category represented by class '0'.

Data pre-processing is a crucial step that involves manipulating data before it is executed, to boost efficiency. It involves data cleaning and data transformation which is seen in the next section after this. To start with data pre-processing, Fig. 1 shows the number of articles in respect of the corresponding class labels fake and real respectively. It is seen that there are not many differences in both datasets. It is clear that the dataset is stabilized. In the figure, the '0' class (in the blue bar) represents the false category of news and '1' (orange bar) represents the true news category. As the contents in the subject section are different in both the categories, the corresponding subject, title and date of each news can be eliminated, thereby keeping only the main text, for further processing.

Fig. 2 explains about the various subjects that together build the dataset. The count of respective subject states the spread of news into the society. The orange coloured bar represents trusted and true news whereas the blue coloured bar shows untrusted or fake news. The topics that real news talks about includes political news and worldwide news. While fake news on the other side speaks in the field of news, politics,

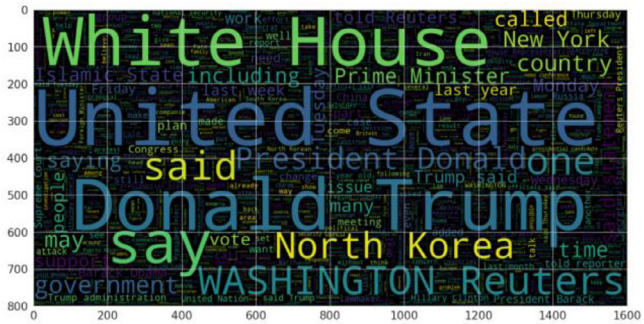

Fig. 3. Representation of Word cloud for original news dataset.

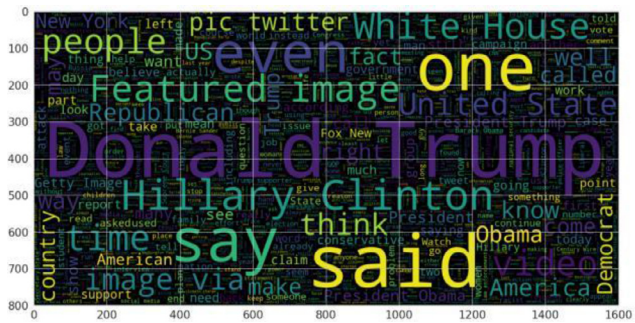

Fig. 4. Representation of Word cloud for fake dataset.

Government news, left news, US news and the news around the middle east region.

4.2.1. Stopwords

Stopwords are English words that don't add much to a sentence's meaning. They can be easily dismissed without compromising the sentence's meaning. Examples for stopwords include words like the, he, and have (Nezu & Miura, 2020). Such words have previously been collected and stored in a package called corpus, which can be observed in the NLTK directory. It can be installed in the Python environment itself.

The Natural Language Toolkit or NLTK is a Python-based collection of tools and applications for symbolic and statistical natural language processing (NLP) for English. To eliminate stop words from a sentence, the text is divided into words and then it is checked to see if the word is in the NLTK list of stop words. If the particular word exists in the collection of a corpus, the word is then eliminated.

4.2.2. Lemmatization

The process of lemmatization is the reduction of inflectional forms-usually meaningful words-to general forms. Stemming refers to the extraction of word stems or root forms that may or may not fully reflect semantic thinking.

4.3. Data processing

Figs. 3 and 4 show the keywords within the set of real news and fake news respectively. To create a word cloud for each category, a maximum of 2000 words are taken into consideration. Such words are known as word clouds that picture bunch of words that are represented and highlighted in a variety of sizes and multiple lengths. In Fig. 3, the louder and larger words observed are White, House, United, State, Donald, Trump which means that these word are equally contributing towards the original news dataset. While in Fig. 4, the bolder and louder words are one, said, Donald, Trump, Hillary, Clinton which means that these words equally contribute in the fake news dataset. The larger and thicker the term, the more frequently it appears in a document and the more significant it is. The words are considered from the input of two separate

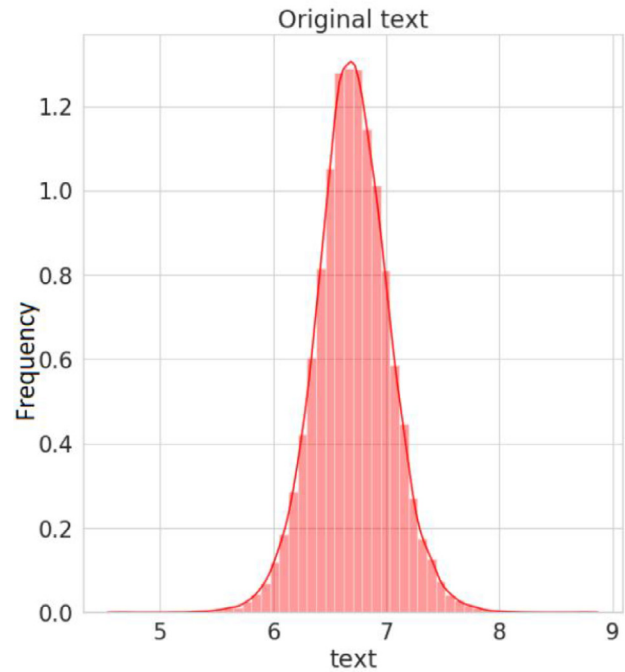

Fig. 5. Length classification of real words in news.

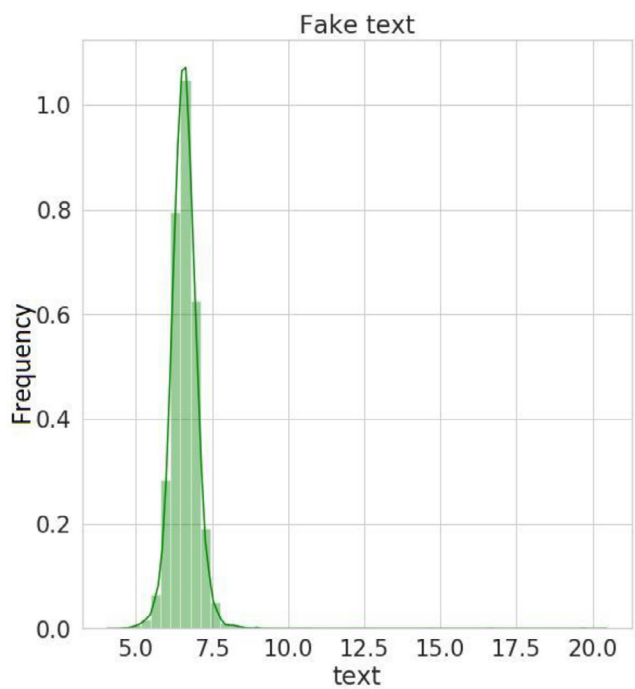

Fig. 6. Length classification of fake words in news.

categories is the one containing fake news and another containing real news. And in accordance with each category, the word cloud is generated.

The Figs. 5 and 6 showcase the data representation of variety of outcomes into with respect to the X-axis. As shown in Fig. 5, the original or real words of the news range from 5 to 8. It can be seen that the texts in original news which range between 6 and 7 have the highest frequency, which means that such texts are most frequent. Whereas in Fig. 6, that is for fake words of the dataset news range in between 5 and 8. It can be seen that the texts in fake news which range between 6 and 7 have the highest frequency, which means that such texts are

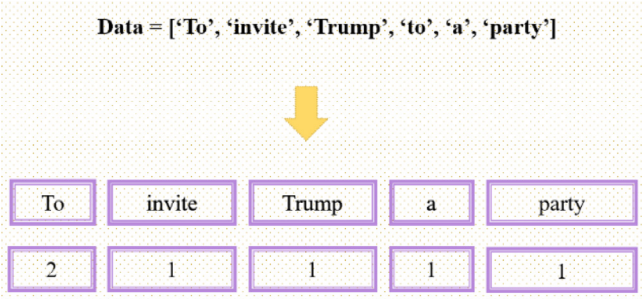

Fig. 7. Process of vectorization.

most frequent. Text data visualisation has several perks, such as quickly determining the most frequently used word to recognise what the text is all about, representing the number of positive and negative reviews in a graph for all data. This kind of data visualization help for further processing as explained in the following sections.

4.4. Tokenization: feature extraction or vectorization

Tokenization is the process of representing each word in the form of a number. For using textual data for predictive modelling, the text is parsed to remove specific words which are known as the process of tokenization. Then, feature extraction (or vectorization) is carried out that is the words are converted to integers or floating-point numbers in order to be used as inputs in machine learning methods.

Machine learning techniques are used extensively in the field of text analysis. However, because most algorithms need numerical feature vectors of a fixed size rather than raw text documents of variable length, the raw data, a sequence of symbols, cannot be provided directly to them. The following are the most frequent methods for extracting numerical features from text content:

- By using white-spaces and punctuation as token separators, text may be tokenized and assign an integer id to each potential token.
- Token occurrences in each document can be counted.
- Tokens that appear in the majority of samples / documents are normalised and weighted with decreasing importance of tokens.

A multidimensional sample is a vector that contains all of the token frequencies for a given document. A matrix having one row each with a document and one column per token (e.g. word) occurring in the corpus can thus be used to represent a corpus of documents. The process of converting a collection of text documents into numerical feature vectors is referred to as vectorization. The Bag of Words or "Bag of n-grams" representation refers to this specific method (tokenization, counting, and normalisation). Word occurrences are used to describe documents, but the relative location information of the words in the document is largely ignored. Here, the CountVectorizer has been used which is a simple tool for tokenizing a collection of text documents as well as building a vocabulary of known words.

CountVectorizer is a fantastic Python tool that is used to convert a given text into a vector-based on the frequency (count) of each word in the full document. This is useful when we have a large number of such texts and want to convert each word in each text into vectors. Fig. 7 shows the process of vectorization word to word in the form of sequence.

Text mining relies heavily on n-grams, which are a set of co-occurring or continuous sequences of n items from a large text or sentence sequence. The item in this case could be words or letters. The unique words in the sentence are referred to as 1 grams or unigrams. Bigram (2-gram) is a two-word combination. A trigram is three words blend. As shown in Fig. 8, from sentence to word by word n-gram analysis has been shown. Unigram is a model that depends solely on the frequency of a word without considering prior words as shown in Fig. 9.

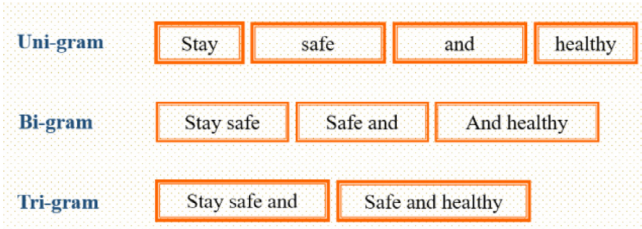

Fig. 8. Example for process for Uni-gram, Bi-gram and Tri-gram.

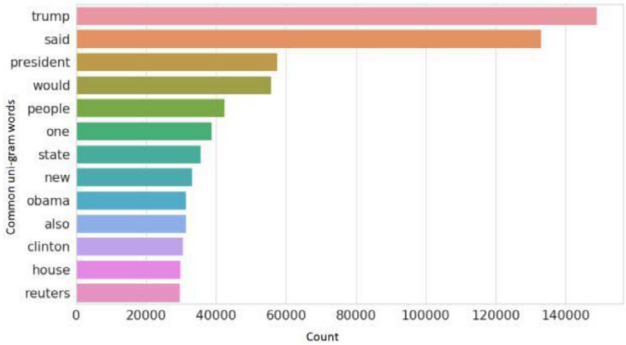

Fig. 9. Uni-gram analysis.

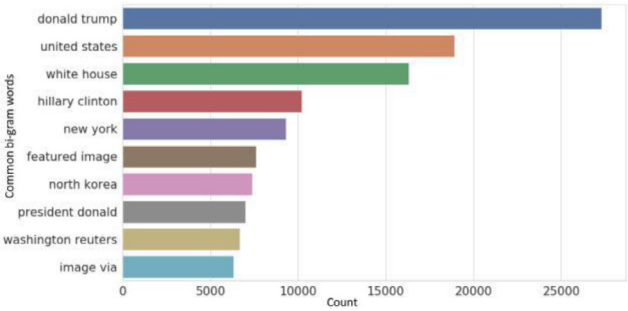

Fig. 10. Bi-gram analysis.

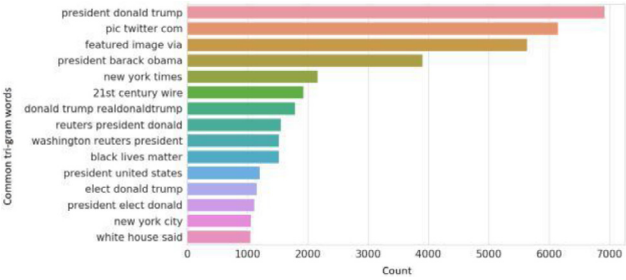

Fig. 11. Tri-gram analysis.

Bigram refers to a model that uses only the prior word to predict the current word as shown in Fig. 10. It's a trigram model if two previous words are taken into account as given in Fig. 11.

An N-gram model predicts the most likely word that could follow a sequence of N-1 words given the sequence. It's a probabilistic model that's been trained on a text corpus. A model like this is useful in a wide range of NLP applications, including speech recognition, machine translation. An N-gram model is constructed by counting the number of times certain word sequences appear in corpus text and then predicting the possibilities. The likelihood of a given N-gram inside any series of words in a language is predicted by an N-gram language model. Authors of this article (Tian, Song, & Xia, 2020) present a character-

based neural model for the joint task that is strengthened by n-gram multi-channel attention in this study. In the attention module, n-gram characteristics are divided into groups based on a variety of parameters, and n-grams within each group are weighted and discriminated according to their importance for the joint task in the given scenario. The social media posts have been classified based on sports filter, where the author has done a comparative analysis between the two feature extraction techniques for sentiment analysis that is, the character n-grams and the word n-grams (Sharounthan, Nawinna, & De Silva, 2021). External self-supervised signals (or knowledge derived through unsupervised learning, such as *n*-grams) have been shown in recent research to be effective in providing meaningful semantic justification for understanding languages like Chinese, as well as improving performance on many downstream tasks. To improve the encoders even further, the authors propose pre-train n-gram-enhanced encoders with a vast volume of data and advanced training methodologies (Song, Zhang, Wang, & Lee, 2021).

The corresponding predicted frequencies for n-grams and (n - 1)-grams may be used to calculate the n-gram probabilities, which is the first essential insight towards that solution. Hence, the frequencies for n-grams and (n - 1)-grams can be:

$$P(w_n|w_1w_2\ldots.w_{n-1}) = \frac{c(w_1\ldots.w_n|L)}{c(w_1\ldots.w_{n-1}|L)} \tag{1}$$

where $c(w|L)$ stands for the predicted number of times the substring w appears in a phrase of L.

4.5. Word embeddings: GloVe

GloVe, as the name, suggests global vectors is an unsupervised learning technique that generates word vector representations. The resulting representations reveal fascinating linear segments of the word vector space, with training based on a corpus of global word to word co-occurrence statistics. The gloVe has an advantage that, unlike Word2vec, it does not solely rely on local statistics (local context information of words) to generate word vectors, but also incorporates worldwide statistics (word co-occurrence).

Existing techniques for learning vector space representations of words have been effective in attracting fine-grained semantic and syntactic regularities using vector arithmetic, but the genesis of these regularities remains a mystery. As a result, a new global log-linear regression model has been developed that combines the benefits of two key model families: global matrix factorization and local context window approach (Pennington et al., 2014).

Because the ratio P_{ik}/P_{jk} is determined by three words, i, j, and k, the most general model is as follows:

$$F(w_i, w_j, w_k) = \frac{P_{ik}}{P_{jk}} \tag{2}$$

where $w \in \mathbb{R}^d$, i and j are words in corpus. The ratio of their co-occurrence probabilities with various probe words, k, can be used to investigate the link between these terms. Where as P_{ik}/P_{jk} is their ratio.

While F may be a sophisticated function specified by a neural network, this would conceal the linear structure which is being attempted to portray. The dot product between the two entities can be considered:

$$F((w_i - w_j)^T w_k) = \frac{P_{ik}}{P_{jk}} \tag{3}$$

The differentiation between a word and a context word in word-word co-occurrence matrices is subjective, and thus swapping is allowed between the two roles. F can hold a homomorphism property which assures that the division F(A)/F(B) can be used to represent the subtraction F(A-B):

$$F(w_i^T w_k) = P_{ik} = \frac{X_{ik}}{X_i} \tag{4}$$

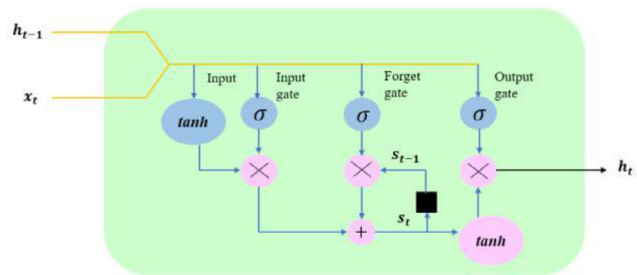

Fig. 12. Long short-term memory architecture.

After the tokenization process, other parameters in scikit-learn can be used to fine-tune the model input. To fine-tune the model input a concept of max features has been used here. Rather than using all words, the model's complexity and size can be reduced by using the maximum number of words. For our model, we considered max features as 10,000 and max length as 300. The GloVe model is trained on the non-zero entries of a global word-word co-occurrence matrix, which keeps track of how often words appear together in a corpus. Collecting the statistics for this matrix necessitates a single run over the full corpus. This pass can be computationally expensive for large corpora, but it is a one-time investment. Because the number of non-zero matrix entries is often substantially fewer than the total number of words in the corpus, subsequent training iterations are substantially faster.

4.6. Neural network: deep learning LSTM model

To build our model, neural networks have been used. And a sequential model is used which is suited for a simple stack of layers with one input tensor and one output tensor for each layer. The model must know what input shape to anticipate. As a result, in a Sequential model, the first layer must receive information on the input shape. Sequence classification is a predictive modelling problem in which the aim is to predict a category given a sequence of inputs that spans space or time. The fact that the sequences might be of varying lengths, contain a broad vocabulary of input symbols and may require the model to learn the long-term context or connections between symbols in the input sequence makes this challenge tough. So, a long short-term memory model has been used to recognize fake news.

An LSTM network is a type of recurrent neural network that uses LSTM cell blocks instead of traditional neural network layers. The input gate, forget gate, and output gate is three components of these cells.

As shown in Fig. 12, the new sequence value x_t is concatenated to the previous output from the column h_{t-1} on the left hand side.

- The first step in suppressing this combined input is to use a tanh layer.
- The input is then routed through an input gate in the second phase. A compressed input is multiplied by the output of an input gate, which is a layer of sigmoid activated nodes. Any parts of the input vector that aren't required can be "killed" by these input gate sigmoids.
- Because a sigmoid function produces values between 0 and 1, the weights connecting the input to these nodes can be taught to produce values near to zero, allowing certain input values to be "switched off."
- The forget gate loop is the next phase in the data flow through this cell. s_t is an forget gate variable in LSTM cells.
- To establish an effective layer of recurrence, s_{t-1} is added to the input data. The use of an addition operation rather than a multiplication operation reduces the possibility of disappearing gradients.
- This repetition loop, however, is regulated by a forget gate, which functions similarly to an input gate but assists the network in learning which state variables should be "kept" or "lost."

- Finally, there's a tanh squashing function at the output layer, whose output is controlled by an output gate. This gate decides which values from the cell h_t are actually allowed as an output.

Using a tanh activation function, the input is squeezed between -1 and 1. This can be represented as follows:

$$g = tanh(b^g + x_t U^g + h_{t-1} V^g) \tag{5}$$

where b^g is input bias, U^g and V^g denote the input and prior cell output weights, respectively.

The forgot gate can be denoted as:

$$f = \sigma(b^f + x_t U^f + h_{t-1} V^f) \tag{6}$$

s_{t-1} is the output of the element-wise product of the previous state and the forget gate.

While, output gate can be given as:

$$o = \sigma(b^o + x_t U^o + h_{t-1} V^o) \tag{7}$$

The final output of the network will be h_t as shown in Fig. 12.

We have used the stacked LSTM model, to enhance our system. While using the stacked version of LSTM, the return sequence is set to true. The output of each neuron's hidden state is used as an input to the following LSTM layer when the return sequence is set to true. To classify a piece of particular news to real news or fake news, a complex LSTM model is required with multiple LSTM and Dense layers.

- The first layer is the Embedding layer, which represents each word with 32 length vectors.
- The next two layers are the LSTM layer, which has 128 and 64 memory units respectively.
- There are two Dense output layers. The first Dense layer consists 32 memory units and reLu activation function.
- The next dense layer is the output layer consisting a single neuron and a sigmoid activation function.

In a neural network, a dense layer is simply a regular layer of neurons. Each neuron in the previous layer receives information from all the neurons in the layer above it, making it tightly linked. A weight matrix W, a bias vector b, and the activations of the previous layer make up this layer. To avoid over-fitting, the authors of this article (Tan et al., 2019) have usually chosen one or two thick layers in their suggested networks.

The Rectifier Unit (ReLu) is the most often used activation function for the outputs of CNN neurons (Li & Yuan, 2017). The key benefit of employing the ReLU function over other activation functions is that it does not simultaneously stimulate all of the neurons. ReLU is a non-linear activation function like tanh or sigmoid that is computed after convolution. ReLu can be denoted as:

$$\sigma = max(0, z) \tag{8}$$

The likelihood of the word j appearing in the context of the word i by assuming that Q_{ij} is sigmoid. The inferred global objective function can be expressed in the following way:

$$J = -\sum_{i \in corpus, j \in context(i)} \log Q_{ij} \tag{9}$$

Because fake news is the classification problem, and we use the Dense output layer for the predictions of two classes that is fake news or real news categorized as 0 and 1 respectively. The optimizer, the loss function and the metrics are meant for the complete compilation of the model.

The model has been trained for 10 epochs. Also, the Binary Cross-Entropy loss function has been implemented and an Adam optimizer is used to upgrade the weights. The learning rate is selected as 0.01. The batch size is set to 256. Whereas the embedding size is 100. To enhance the accuracy of our model, we have reduced the batch size. The proposed model consists of all 168,769 trainable parameters. The training of the proposed model is built using the Windows machine with an AMD Threadripper processor of 64 cores, 128 threads and 256 MB cache. The graphic card used was RTX3080 and RAM of about 128 GB.

Table 1
LSTM layered architecture.

Layer (type)	Output size	Param number
Embedding_1 (embedding)	300 x 100	1,000,000
Lstm_1 (LSTM)	300 x 128	117,248
Lstm_2 (LSTM)	64	49,408
Dense_1 (Dense)	32	2080
Dense_2 (Dense)	1	33

Table 2
Hyperparameters for proposed model.

Hyperparameters	Value
Embedding layer	1
LSTM layer	2
Dense layer	2
Loss Function	Binary cross entropy
Activation function	ReLu
Optimizer	Adam
Learning rate	0.01
Number of epochs	10
Embedding size	100
Batch size	256

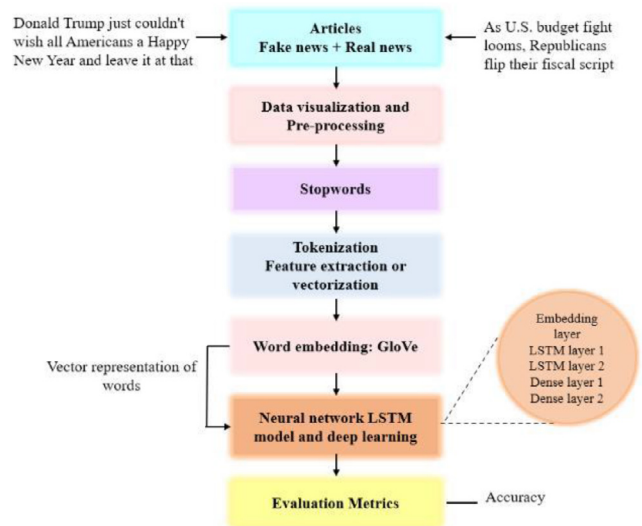

Fig. 13. Model architecture of proposed system using LSTM neural networks.

5. Results

The model has now been trained to recognize fake news as well as real news. The proposed model is evaluated based on accuracy, precision, recall, F1-score and support metrics. Additionally, a comparative analysis is carried out as shown in Table 3. In this table, a comparative analysis is shown along with the model accuracy and year of publication.

Testing and training is a very essential part while building any model. The complete base of the model is relied on how far we train the model. Fig. 14 shows testing and training accuracy details obtained concerning the number of epochs. As the model is trained with 10 number of epochs, it is seen from the figure that the training and testing accuracy increases until the maximum epochs are considered. Whereas Fig. 15 demonstrates testing and training loss details obtained with respect to the number of epochs. From the obtained graph it is clear that the training and testing loss is getting decreased with a number of epochs, that is minimum loss is met.

Fig. 16 is the confusion matrix obtained for the proposed model. It shows that the proposed model achieved 99.88% accuracy.

Table 3
Comparative analysis with various research work.

Sr.No	Title	Accuracy	Year
1	FakeBERT: Fake news detection in social media with a BERT-based deep learning approach	98.90%	2021
2	DeepFakE: improving fake news detection using tensor decomposition-based deep neural network	88.64%	2020
3	EchoFakeD: improving fake news detection in social media with an efficient deep neural network	92.30%	2021
4	Adaptive Salp swarm optimization algorithms with inertia weights for novel fake news detection model in online social media	99.50%	2021
5	ConvNet frameworks for multi-modal fake news detection	93.56%	2021
6	**Our Method**	**99.88%**	

Fig. 14. Testing and training accuracy representation against number of epochs.

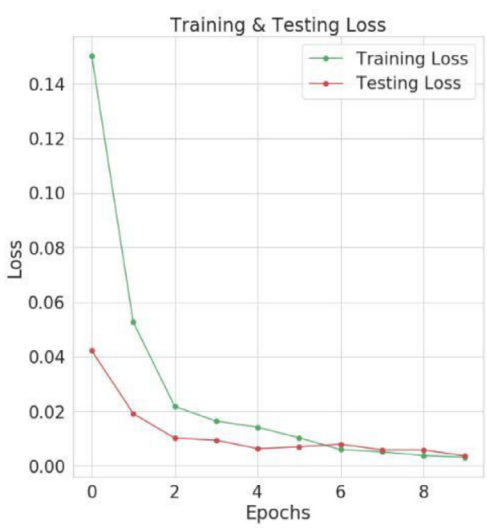

Fig. 15. Testing and training loss representation against number of epochs.

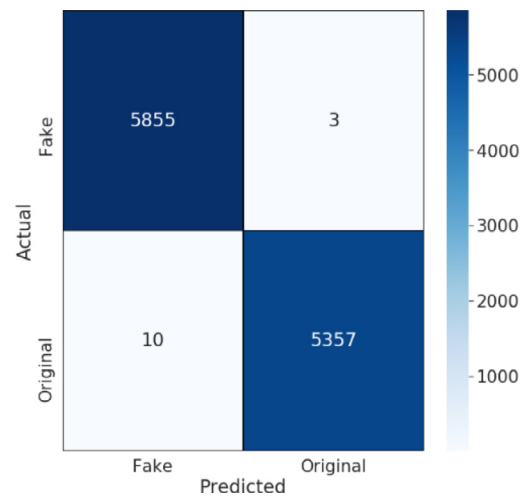

Fig. 16. Confusion matrix for proposed model.

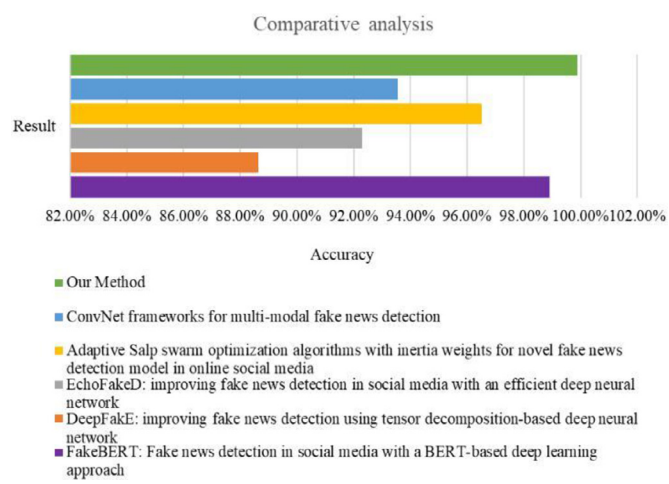

Fig. 17. Comparative analysis of proposed model with other models.

As shown in Fig. 17, a comparative approach has been analyzed among the proposed method and other methods for fake news detection. The accuracy and methods are compared for the analysis. The graph in all shows six comparative methods including the proposed system. The details of the same graph are described in Table 3. From the graph, it can be seen that the proposed system outperforms with 99.88% of accuracy as compared to other methods, followed by FakeBERT which achieved 98.90%.

6. Discussion

6.1. Theoretical contributions

The discovered information can assist researchers in doing primary research. This research gains advantage in identifying the fake news from several overall original news. It benefits society as well as the government from getting into the suspect of fake news. The main theoretical concepts used to build the model to detect fake news includes the LSTM architecture and a deep learning-based model. Also, the main concept to build the model included the pre-trained word embedding process, that is Global Vectors for Word Representation GloVe. Also, LSTM has been used to enhance to the proposed model. Long Short-Term Memory

(LSTM) is a recurrent neural network with a tree structure that is used to assess variable-length sequential input (Bahad, Saxena, & Kamal, 2019).

6.2. Implications for practice

Fake news is in the trend of the ongoing century. The more social media applications, the more is the possibility of spreading fake news using civilization as a tool. Researchers and educators will benefit from this study since it will provide a comprehensive analysis of the literature. Journals will gain a better grasp of the most widely expanded topics in the world of fake news, as well as a clearer understanding of fake news propagation, which will assist our society and funding organization as well as the government in making informed decisions about further plans. Thoughts and data posted on the internet and other social media platforms affect public opinion about possible treatments for detecting and treating illnesses. This influences people' health as well as medical experts, who are forced to defend their diagnosis as well as the therapies they recommend against misguided people. An algorithm is proposed to analyse the behaviour of Twitter users. The suggested analytical scheme is built on the interaction between the computer tool where the correct question needs to be asked to the tool in order to follow the line of any suspect data (Lara-Navarra, Falciani, Sánchez-Pérez, & Ferrer-Sapena, 2020). Besides social media, it is also widely used in a similar manner in political domains as well as healthcare sectors, various media websites.

7. Conclusion and future work

The focal point of our research lies in differentiating and detecting fake news and original or real news. The goal of this study is to determine and comprehend the impact of fake news on the government and society. This research aims in helping society from the spread of fake news, making people aware of the fake news propagators and their expansion in today's world. As a result, the spread of fake news has been a hot topic for quite some time. The primary obligation for the betterment of society is to detect fake news. The proposed model used a deep learning framework, which makes use of neural networks and the long short-term memory architecture. This neural network model has been presented to improve the detection of misleading news. Before training the model, the concept of stopwords has been employed for data preprocessing, which enhances the accuracy of the model. A tokenization method has been proposed for feature extraction or vectorization, which assigns tokens to word embeddings. The GloVe technique was used to represent each word in vector form for word embeddings. The information then passes via the main LSTM neural network, where it is routed through various levels in the architecture. The long short -term memory model is used for the construction of the model which mainly is based on the concept of classification of data for sequential prediction. Finally, the model is programmed to distinguish between true and false news. The evaluation metric of the proposed model is accuracy. Our proposed model achieved 99.88% accuracy. The future direction lies to improvise and outstretch the existing work for implementation of existing work towards building an automated system for e-commerce websites, where detection of fake news has become equally important.

8. Declaration of Competing Interest

The author declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Tavishee Chauhan: Conceptualization, Methodology, Software. **Hemant Palivela:** Conceptualization, Methodology, Software.

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